GGC-P-00475(GSA-FSS)
June 1, 1964

INTERIM FEDERAL SPECIFICATION

PLIERS, JEWELERS

This Interim Federal Specification was developed by the General Services Administration, Federal Supply Service, Standardization Division, Washington, D. C. 20407, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above.

1. SCOPE AND CLASSIFICATION

1.1 Scope. - This specification covers pliers used by jewelers for shaping the inside coil of a watch mainspring, and for forming pendant bows and rings.

1.1.1 Federal specification coverage. - Federal specifications do not include all varieties of the commodity indicated by the titles of the specifications, but cover only those generally used by the Federal Government.

1.2 Classification. - The jewelers' pliers covered by this specification shall be of the following types:

Type I - Mainspring jewelers pliers.
Type II - Pendant bow contracting style jewelers pliers.

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards. - The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification:

Federal Standards:
Fed. Std. No. 102 - Preservation, Packaging and Packing Levels.
Fed. Std. No. 123 - Marking for Domestic Shipment (Civilian Agencies).
Fed. Std. No. 151 - Metals; Testing Methods.

Military Specifications:
MIL-H-15424 - Hand Tools; Packaging Of.

Military Standards:
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

FSC 5120

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2.2 Other publications. - The following publications, of the issue in effect on date of invitation for bid, forms a part of this specification:

Governmental:
National Bureau of Standards Handbook:
H-28 - Screw-Thread Standard for Federal Services.

(Copies of National Bureau of Standards Handbook may be obtained upon application, accompanied by money order, coupon, or cash to the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402. Prices may be obtained from the Superintendent of Documents.)

3. REQUIREMENTS

3.1 Illustrations. - The illustrations shown herein are for the convenience of identification, and are not intended to preclude the purchase of jewelers pliers which are otherwise in accordance with the requirements of this specification.

3.2 Material. - The metals used in the pliers shall be of such composition and quality that the pliers can meet the physical requirements of this specification. The metals shall be free of defects that would impair the usefulness or durability of the pliers.

3.3 Design. - Pliers of the type specified shall be similar in overall design to figure 1 or 2 as applicable.

3.3.1 Joints. - Pliers shall have a box type joint. The stud shall have a Rockwell hardness of "C" 50-60.

3.3.2 Threads. - Threads in or on component parts of pliers shall conform to requirements in the National Bureau of Standards Handbook H-28.

3.3.3 Handle. - The handle of the pliers shall be so designed as to afford a comfortable hand grip. Tips of handles shall not touch each other when the jaws are in a closed position. The degree of tightness or looseness of the handles on the stud, shall be such that there is no binding or play in any direction. The Rockwell hardness of the handles shall be "C" 40-50.

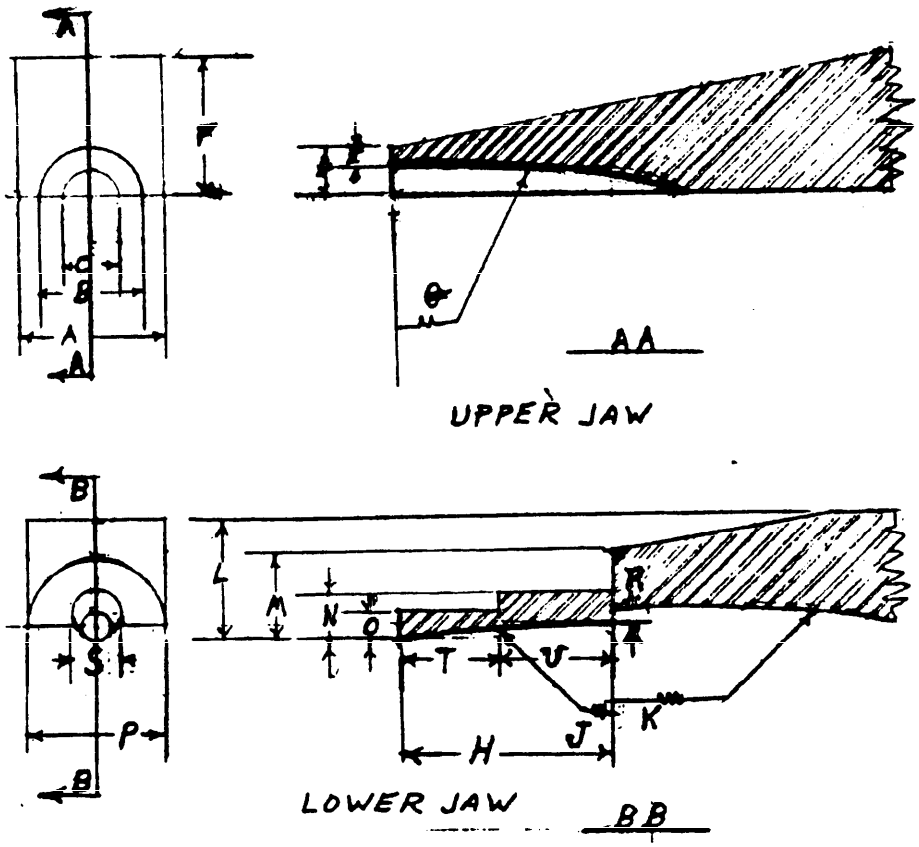
3.3.3.1 Deflection and permanent set. - The pliers shall not exceed the deflection and permanent set specified in table I when tested in accordance with 4.5.3.

Table I. - Allowable deflection and permanent set for pliers.

Type	Thickness of metal between jaws	Test load	Deflection under load max.	Permanent set max.
	Inch	Inch Pounds	Inch	Inch
I	0.010 ± 0.005	100	.50	.03
II	0.062 ± 0.005	400	.94	.03

3.3.4 Type I. - The pliers shall be so designed that they will easily shape the inside coil of a watch mainspring to fit on a watch mainspring barrel arbor. The dimensions of the pliers shall conform to table II requirements. The jaws shall have a Rockwell "C" hardness of 40-50. The minimum jaw opening, measured at the tip of the jaws in a completely open position, shall be 1-inch. The clearance between the upper and lower jaws, taken at the back end of the jaws, shall be 0.031 ± 0.005 inch when the jaws are in a closed position. Length overall shall be 4-1/2 inches ± 1/8 inch.

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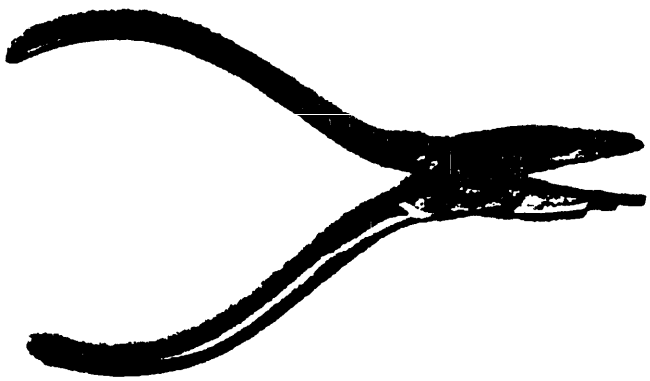


Figure 1. - Type I.

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Table II. - Type I jewelers pliers.

Dimension Letter	Dimension Inches	Tolerance Inches	Dimension Letter	Dimension Inches	Tolerance Inches
A	0.270	± 0.010	N	0.125	± 0.010
B	0.185	± 0.010	O	0.085	± 0.010
C	0.085	± 0.010	P	0.275	± 0.010
D	0.075	± 0.010	R	0.031	± 0.010
E	0.035	± 0.010	S	0.130	± 0.010
F	0.225	± 0.010	T	0.186	± 0.010
G	2 R	± 1/64	U	0.186	± 0.010
H	0.372	± 0.010	-	-----	-----
J	3 R	± 1/64	-	-----	-----
K	3 R	± 1/64	-	-----	------
L	0.250	± 0.010	-	-------	------
M	0.200	± 0.010	-	-------	------

3.3.5 Type II. - The pliers shall be so designed that they will easily form a watch pendant or finger ring to the desired shape. Dimensions shall conform to table III requirements. The female jaw insert shall be made of extra hard brass having a Rockwell "B" hardness of 90-96. The male jaw and the female jaw support shall have a Rockwell "C" hardness of 40-50. The minimum jaw opening measured at the tip of the jaws, in a completely open position, shall be 2-inches. The pliers shall be provided with a stop to keep the jaws from closing. The clearance between the jaws in a closed position shall be 0.032 at the tips and 0.071 at the back ± 0.005 inch. Length overall shall be 6-1/4 inches ± 1/8 inch.

Table III. - Type II, Jewelers' Pliers.

Dimension Letter	Dimension Inches	Tolerance Inches	Dimension Letter	Dimension Inches	Tolerance Inches
A	0.090	± 0.010	K	0.133	± 0.010
B	0.305	± 0.010	L	0.100	± 0.010
C	0.485	± 0.010	M	1.125	± 0.010
D	0.125	± 0.010	N	0.425	± 0.010
E	0.169	± 0.010	O	0.270	± 0.010
F	0.090	± 0.010	P	-----	-------
G	0.545	± 0.010	R	-----	-------
H	0.725	± 0.010	S	-----	-------
J	1.135	± 0.010	T	-----	-------

3.4 Construction.- Pliers shall be so constructed as to withstand tests in section 4 without distortion.

3.5 Finish.- All surfaces shall be free of pits, nodules, flash marks, burrs, cracks or scale. Visible surfaces, other than jaws, shall have a surface smoothness of 63 microinches or better. The Jaws shall have a surface smoothness of 32 microinches or better. All bearing surfaces shall be sufficiently smooth to resist wear and not cause binding.

3.5.1 Coating.- Type II pliers shall be electroplated with nickel or chromium as specified (see 6.1). Plating shall completely cover all visible surfaces.

3.6 Marking.- The pliers shall be marked in a plain and permanent manner with the manufacturer's name or with a trademark of such known character that the source of manufacture may be readily determined. Pliers shall be marked before plating.

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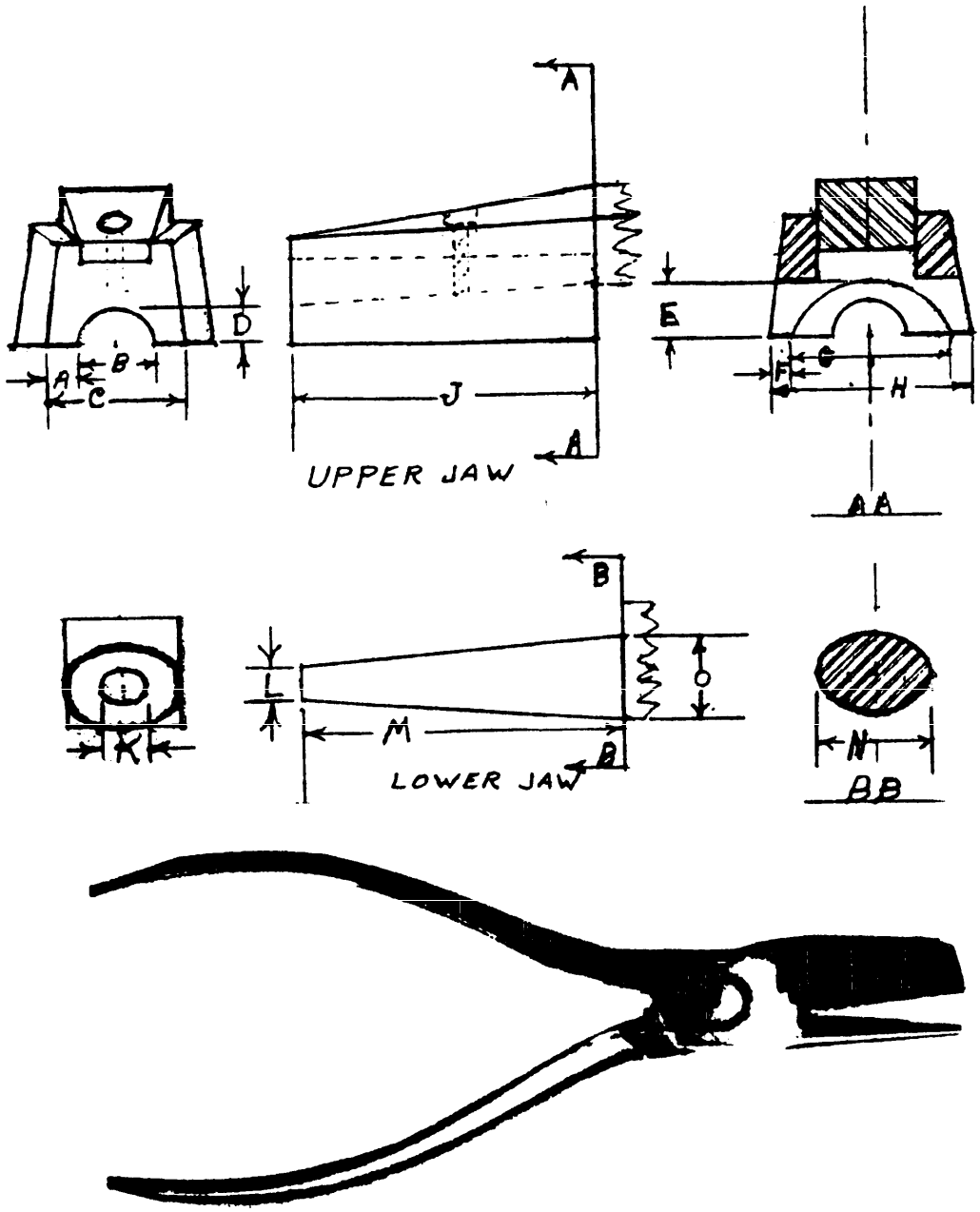


Figure 2. - Type II

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3.7 Workmanship.- The workmanship shall be first class in every respect.

4. SAMPLING, INSPECTION AND TEST PROCEDURES

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Inspection of materials and components.- In accordance with 4.1 above, the supplier is responsible for insuring that materials and components used were manufactured, tested and inspected in accordance with the requirements of referenced subsidiary specifications and standards to the extent specified herein, or, if none, in accordance with this specification.

4.2 Sampling for inspection.- Sampling shall be performed in accordance with MIL-STD-105.

4.2.1 Sample unit and lot formation.- The sample unit shall be one plier of one type. The lot shall be comprised of pliers of one type manufactured by one supplier and submitted for inspection at one time.

4.2.2 Sampling for visual examination.- Each sample unit selected in accordance with 4.2 shall be examined as stated in 4.3. The inspection level shall be level II with an AQL of 2.5 defects per hundred units.

4.2.3 Sampling for dimensional examination.- Each sample unit selected in accordance with 4.2 shall be examined as stated in 4.4. The inspection level shall be S-3 with an AQL of 2.5 percent defective.

4.2.4 Sampling for test.- Each sample unit selected in accordance with 4.2 shall be tested in accordance with 4.5. The inspection level shall be level S-3 with an AQL of 1.5 percent defective.

4.3 Visual examination.- Each sample unit shall be examined to determine compliance with the visual requirements of this specification for material, design, construction, finish and workmanship.

4.4 Dimensional examination.- Each sample unit shall be examined to determine compliance with dimensional requirements specified herein. Any deviation shall constitute a defect.

4.5 Testing.- Each sample unit shall be tested in accordance with 4.5.1 and 4.5.2 to determine compliance with applicable section 3 requirements.

4.5.1 Handle load test.- The sample plier under test shall be supported in a manner suitable for applying the load properly and determining the deflection and permanent set of the handles. A platform scale similar to figure 3 may be used in the performance of the load test. The gripping surfaces of the jaws shall be placed in a horizontal position, with the head resting on a support or guide. The lower handle shall rest on a suitable fixed knife-edge support at the point of maximum handle curvature (the center of handle bow). Loads shall be applied through a knife edge to the upper handle at the point of maximum handle curvature (the center of the bow) and in line with the lower handle support. The test load applied to the handle shall be calculated by dividing the test load specified herein by the distance between the center of the stud and the line of action of the load. Steel inserts used in connection with the test shall have a Rockwell "C" number of not less than 20. The width of the flat steel insert shall be 1/16 inch ± 1/64 inch. The method of determining the amount of deflection under load and permanent set shall be as follows: Mark the extreme ends of the handles with shallow center punch marks. Place pliers on test fixture and apply a five pound load. Take a measurement of distance between center punch marks. Apply specified test load and take a measurement. This measurement is the amount of deflection. Reduce the test load to five pounds and take a measurement. This measurement is the amount of permanent set. The load and deflection test may be performed on testing equipment other than that specified herein provided the same results will be obtained. If the deflection or permanent set is in excess of that specified in table I, or the plier is deformed to the extent that it will not perform its intended function, it is a defective. Tips of handles may be ground off for placing of center punch marks.

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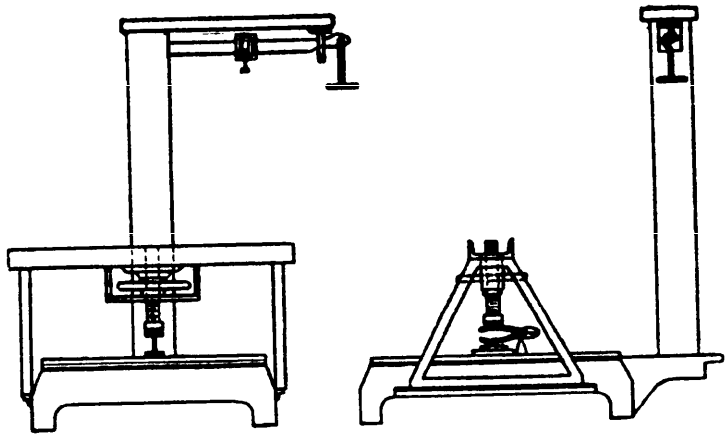


Figure 3. - Apparatus for applying test load to handles of pliers.

4.5.2 Hardness test.- Each plier in the sample shall be tested for hardness specified herein. One reading shall be taken on each handle and one reading shall be taken on each jaw or jaw insert. Testing shall be performed in accordance with requirements for Rockwell testing (Method 243.1) in Fed. Std. No. 151.

4.6 Inspection of preparation for delivery.- Inspection shall be performed in accordance with requirements of MIL-H-15424 for pliers thru 6 inches.

5. PREPARATION FOR DELIVERY
(Civil agencies. - Fed. Std. No. 102 should be referred to for definitions and applications of various levels of packaging protection for supplies and equipment.)

5.1 Preservation and packaging.- See 6.1(d).

5.1.1 Level A.- Level A cleaning, preservation, and packaging shall be in accordance with level A of MIL-H-15424 for pliers thru 6 inches.

5.1.2 Level C.- Preservation and packaging which will afford adequate protection against corrosion, deterioration, and damage during shipment from supply sources to the first receiving activity for immediate use (vendor to user).

5.2 Packing.- See 6.1(d).

5.2.1 Level A.- Level A packing shall be in accordance with level A of MIL-H-15424 for pliers thru 6 inches.

5.2.2 Level B.- Level B packing shall be in accordance with level B of MIL-H-15424 for pliers thru 6 inches.

5.2.3 Level C.- Packing which will afford protection against damage during direct shipment from the supply source to the first receiving activity for immediate use.

5.3 Marking.

5.3.1 Civil agencies.- In addition to any special marking specified in the contract or order, each unit and intermediate package and shipping container shall be marked in accordance with Fed. Std. No. 123.

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5.3.2 Military.- Interior packages and shipping containers shall be marked in accordance with Military Standard MIL-STD-129.

6. NOTES

6.1 Ordering data.- Purchasers should exercise any desired option offered herein and procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type of pliers required (see 1.2).
(c) Specify metal to be used for plating (see 3.5.1).
(d) Level of preservation, packaging and packing required (see 5.1 and 5.2).
(e) Special marking requirements (see 5.3.1).

GSA DC 88-978

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